

3 compound propositions

1. Always true - tautology
2. Never true - contradiction
3. Neither a tautology nor contradiction - contingency

TABLE 2 De Morgan's Laws.

$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

propositions worth remembering

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \Rightarrow q \equiv \neg p \vee q$$

$$(p \Rightarrow q) \wedge (q \Rightarrow p) \equiv p \Leftrightarrow q$$

$$\neg(p \Rightarrow q) \equiv p \wedge \neg q$$

Quantification Over Restricted Domains:

Let $Q(x)$ be $2x > x+4$

Then $Q(2) \wedge Q(3) \wedge Q(4) \equiv T$ students in class example:

$$\forall x \in \mathbb{Z}, x \geq 5: Q(x)$$

$$\forall x \in \mathbb{Z}: (x \geq 5 \Rightarrow Q(x))$$

$$\exists x \in \mathbb{Z}: (x \geq 5 \wedge Q(x))$$

$$\exists! x \in \mathbb{Z}: (x \geq 5 \wedge Q(x))$$

$$\begin{aligned} \neg \forall x \in S: P(x) &\equiv \exists x \in S: \neg P(x) \\ \neg (P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots) \\ &\equiv \neg P(x_1) \wedge \neg P(x_2) \wedge \neg P(x_3) \wedge \dots \\ &\equiv \exists x \in S: \neg P(x) \end{aligned}$$

$$\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}: P(x, y) = x + y = 0 \equiv T$$

$$\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}: P(x, y) = F$$

Every 15 seconds, someone steals car

Open propositions (propositional function)

$$f(x) = 2x \times x$$

$$f(-1) = -2 > 1 \equiv F$$

$$f(0) = 0 > 0 \equiv F$$

$$f(1) = 2 > 1 \equiv T$$

$$f(1) \wedge f(2) \wedge f(3) \wedge \dots \equiv T$$

$$\forall x \in \mathbb{Z}^+: f(x)$$

Quantifiers:

$\forall x \in \mathbb{N}$: for all in $\mathbb{N}$

$\exists x \in \mathbb{N}$: for at least 1 in $\mathbb{N}$

$\exists! x \in \mathbb{N}$: for just 1 in $\mathbb{N}$

Preference order:

$$\forall x \in \mathbb{Z}: x \leq 3 \Leftrightarrow 4 > 5$$

$$\boxed{\forall x \in \mathbb{Z}: x \leq 3 \Leftrightarrow 4 > 5}$$

F

$\equiv T$

$$\forall x \in \mathbb{Z}: (x \leq 3 \Leftrightarrow 4 > 5)$$

F

Compound propositions

$\neg$ = NOT, $\vee$ = OR, $\wedge$ = AND, $\Rightarrow$ = IMPLIES, $\Leftrightarrow$ = EQUIVALENT

$\wedge$ = and (1st) Order of operations

$\vee$ = or

$\Rightarrow$ = if p, then q | $p \Rightarrow q$ | train ticket example | p =

assumption | q = consequence

The table shows the truth values for compound propositions. The first column lists the truth values for p and q. The second column shows the truth value for $p \vee q$. The third column shows the truth value for $p \wedge q$. The fourth column shows the truth value for $p \Rightarrow q$. The fifth column shows the truth value for $p \Leftrightarrow q$.

p	q	$p \vee q$	$p \wedge q$	$p \Rightarrow q$	$p \Leftrightarrow q$
T	T	T	T	T	T
T	F	T	F	F	F
F	T	T	F	T	F
F	F	F	F	T	T

$\Leftrightarrow$ = p if and only if q | $p \Leftrightarrow q \Leftrightarrow p \Rightarrow q \wedge q \Rightarrow p$

$\oplus$ = XOR | either p or q

The Precedence Hierarchy of Logical Operations

$\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow, \oplus$

brug alligevel altid parentes

p	q	r	$p \vee q$	$p \vee r$	$q \wedge r$	$(p \vee q) \wedge (p \vee r)$	$p \vee (q \wedge r)$
T	T	T	T	T	T	T	T
T	F	T	T	T	F	T	T
T	T	F	T	T	F	T	T
T	F	F	T	T	F	T	T
F	T	T	T	T	T	T	T
F	F	T	F	T	F	F	F
F	T	F	T	F	F	F	F
F	F	F	F	F	F	F	F